WAC 4.0: Windows servers, the directory and websites

Aerospace Cyber Defence Centre of Excellence (CEC), French Air and Space Force Academy

The articles on versions 2.0 and 3.0 presented WAC (Windows Artefacts Collector): a free collector, in C++, that reads volumes raw, writes nothing on the examined machine, and covers workstations from XP SP1 to Windows 11. Version 4.0 moves on to servers: every generation from Server 2003 to 2025, the directories of domain controllers, group policies, web servers and the DNS, DHCP and Hyper-V roles.

Every server, from 2003 to 2025

The same executables collect Server 2003, 2008 R2, 2012, 2012 R2, 2016, 2019, 2022 and 2025, as well as Server 2022 as a Core installation. Each passed the test cycle on the server itself, and that pass found what a workstation did not show: a final LZX block of odd size that left `shell32.dll` half read, older formats of Defender, and the ShimCache of Server 2003, now decoded. Server 2008 without R2 is not measured, for want of an official image. The cycles run as in the field: WAC is started from a USB key and writes there, never on the examined disk.

The directory, without its secrets

On a domain controller, `ntds.dit` and its logs are copied by raw reading, with neither LDAP nor replication; the logs are replayed into the working copy, because recent objects are only there. From it WAC draws each object, deleted ones included, each value of a linked attribute with its removal date (who entered a group, and who left), the replication metadata attribute by attribute, and the security descriptors, where the rights of a DCSync attack can be read. **Secrets are neither read nor written**: only a closed list of attributes is decoded, and an attribute unknown to the schema stays unread. The copy of `ntds.dit` holds them encrypted, and is protected as a hive is.

Group policies

`SYSVOL` is read the same way. For each policy: the registry values of `Registry.pol`, scripts, scheduled tasks of the preferences, security template. A password stored in a preference (`cpassword`) is flagged by its file; its value is not written.

Web servers

IIS, Apache and nginx are found through their configuration and services: sites, pools, modules, access and error logs, in UTC. Each site's folder is copied whole, with any webshell it may hold. The connection strings of `web.config` point to the databases to take; the accounts of SQLite databases are drawn from them with the scheme of their hash (phpass, bcrypt, ASP.NET Identity), and the data folders of MySQL, MariaDB and PostgreSQL are collected. Not read: `wp-config.php` and `.env`.

Roles and accesses

DNS (settings, plug-in DLLs among them, zones, records, debug log), DHCP (scopes and leases of `dhcp.mdb`, audit logs) and Hyper-V (machines and the paths of their disks) are confronted with what each role says of itself. User Access Logging (`LogFiles\Sum`), which a comparison with KAPE showed missing, tells by role, account and address who reached the server, and on which days; its databases are confronted with `dissect.esedb`.

Checked against the live directory

On a Server 2022 controller, the 3 711 objects WAC returns are identical to those of `Get-ADObject`. WAC's ESE reader, set against `dissect.esedb` on the same `ntds.dit`, gives 145 421 values without a single difference. Policies are checked against a witness policy, the controller's events against `Get-WinEvent`. A replication request through DirSync leaves a 4662 event for right `1131f6aa`, found in the collection.

What remains to be measured

The controller was tested only under Server 2022, alone in its forest, with no real `sIDHistory`. Right `1131f6ad` of a true DCSync has no witness. Server 2003's directory format is copied, not converted. On the controller, Windows does not update access dates: the witness check of harmlessness proves only the USN number there. The comparison with KAPE under Sysmon is still the workstation one of 3.0.

In figures

Artefacts collected (3.0: 80)	87
Server generations measured	2003 to 2025
Checks in the cycle, domain controller	609
Checks in the cycle, Windows 11	605
Directory objects identical to `Get-ADObject`	3 711
`ntds.dit` values identical to dissect	145 421
Directory secrets read or written	0

Availability

Version 4.0.0, Windows XP SP1 to 11 and Server 2003 to 2025 (x86 and x64), GPL 3.0 licence: `https://github.com/dev-cec/WAC`. Manuals, validation, harmlessness and comparison files in French and English; server cycles from 31 October to 8 November 2026.



WAC collecting.